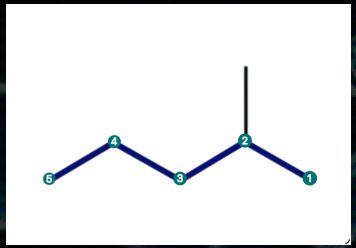
CHEM 13 ACTIVITY 3: NAME IT, DRAW IT

LAGUILLES, Rikki Kirsten M.
BSChE-2A

CARD A01 • 2-METHYLPENTANE

LEVEL: FOUNDATION **MOLECULAR FORMULA: C_6H_{14}**

1. NUMBERED PARENT



2. NAMING TAIL

Parent: Pentane (5 C)

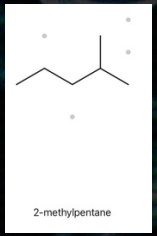
Numbering: Right to left; methyl gets 2

Substituents: Methyl at C2

3. MY NAME

2-METHYLPENTANE

4. THE ALKANE CHECK



Returned name: 2-methylpentane

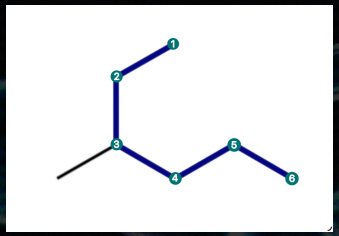
✓ **Match**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A02 • 3-METHYLHEXANE

LEVEL: FOUNDATION **MOLECULAR FORMULA: C_7H_{16}**

1. NUMBERED PARENT



2. NAMING TAIL

Parent: Hexane (6 C), through upper endpoint, junction, and lower zigzag to right endpoint

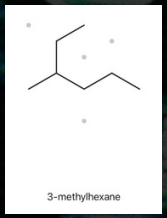
Numbering: Upper endpoint to right endpoint; methyl 3 gets 3 (reverse gives 4)

Substituent: Methyl at C3

3. MY NAME

3-METHYLHEXANE

4. THE ALKANE CHECK



Returned name: 3-methylhexane

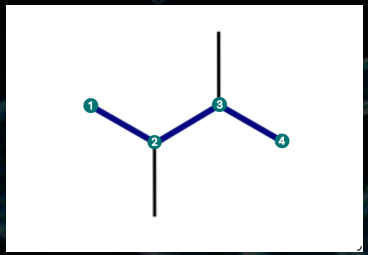
Match

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A03 • 2,3-DIMETHYLBUTANE

LEVEL: FOUNDATION MOLECULAR FORMULA: C_6H_{14}

1. NUMBERED PARENT



2. NAMING TAIL

Parent: Butane (4 C), through both central junctions

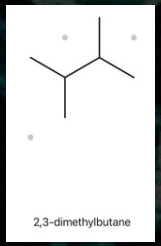
Numbering: Either directions gives 2,3

Substituent: Methyl at C2 and C3; use di-

3. MY NAME

2,3-DIMETHYLBUTANE

4. THE ALKANE CHECK



Returned name: 2,3-dimethylbutane

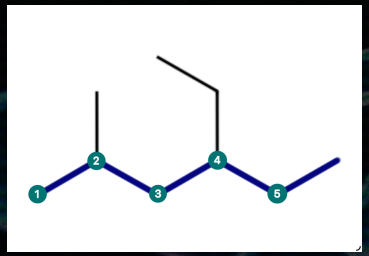
Match

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A04 • 4-EHTYL-2-METHYLHEXANE

LEVEL: DEVELOPING **MOLECULAR FORMULA: C_9H_{20}**

1. NUMBERED PARENT



2. NAMING TAIL

Parent: Hexane (6 C), from left terminal through lower zigzag to upper terminal

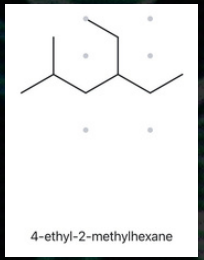
Numbering: Chosen direction gives 2,4; revenge gives 3,5

Substituent: Methyl at C2; ethyl at C4

3. MY NAME

4-EHTYL-2-METHYL HEXANE

4. THE ALKANE CHECK



Returned name: 4-ethyl-2-methylhexane.

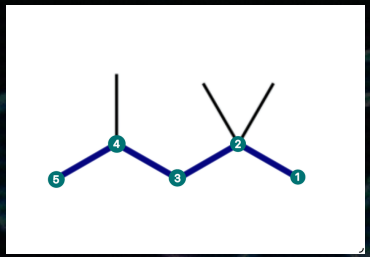
Match

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A05 • 2,2,4-TRIMETHYLPENTANE

LEVEL: DEVELOPING **MOLECULAR FORMULA: C_8H_{18}**

1. NUMBERED PARENT



2. NAMING TAIL

Parent: Pentane (5 C), through the highly branched carbon

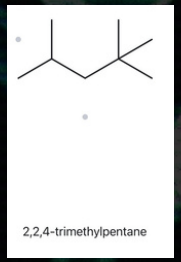
Numbering: Gives 2,2,4; reverse gives 2,4,4

Substituent: Methyl x2 at C2 and methyl x1 at C4; use tri-

3. MY NAME

2,2,4-TRIMETHYL PENTANE

4. THE ALKANE CHECK



Returned name: 2,2,4-trimethylpentane

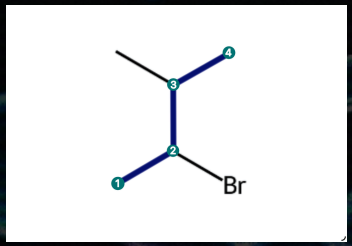
✓ **Match**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A06 • 2-BROMO-3-METHYLBUTANE

LEVEL: DEVELOPING **MOLECULAR FORMULA: $C_5H_{11}Br$**

1. NUMBERED PARENT



2. NAMING TAIL

Parent: Butane (4 C)

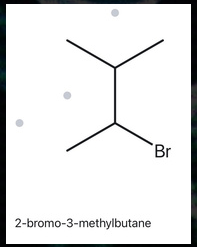
Numbering: Both direction gives 2,3; alphabetical tie-brake gives bromo the lower locant

Substituent: Bromo at C2; methyl at C3

3. MY NAME

2-BROMO-3-METHYL BUTANE

4. THE ALKANE CHECK



Returned name: 2-bromo-3methylbutane

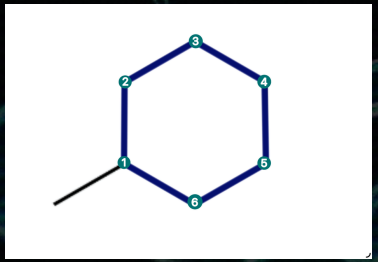
✓ **Match**

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A07 • METHYLCYCLOHEXANE

LEVEL: DEVELOPING **MOLECULAR FORMULA: C_7H_{14}**

1. NUMBERED PARENT



2. NAMING TAIL

Parent: Cyclohexane ring (6 C)

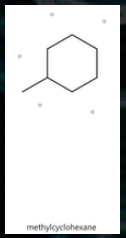
Numbering: Substituted ring carbon is C1; locant is 1 omitted for a single substituent

Substituent: Methyl at C1

3. MY NAME

METHYL CYCLOHEXANE

4. THE ALKANE CHECK



Returned name: Methylcyclohexane

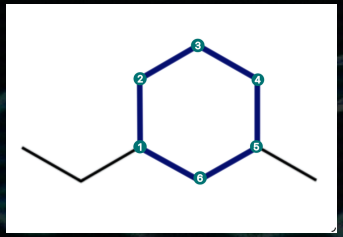
Match

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A08 • 1-ETHYL-3-METHYLCYCLOHEXANE

LEVEL: CHALLENGE **MOLECULAR FORMULA: C_9H_{18}**

1. NUMBERED PARENT



2. NAMING TAIL

Parent: Cyclohexane ring (6 C)

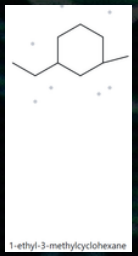
Numbering: Either direction gives 1,3; tie-brake assigns C1 to ethyl

Substituent: Ethyl at C1; methyl at C3

3. MY NAME

1-ETHYL-3-METHYL CYCLOHEXANE

4. THE ALKANE CHECK



Returned name: 1-ethyl-3-methylcyclohexane

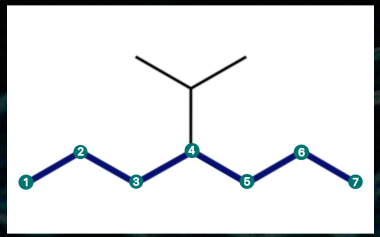
✓ Match

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A09 • 2-(PROPAN-2-YL)HEPTANE

LEVEL: CHALLENGE **MOLECULAR FORMULA: $C_{10}H_{22}$**

1. NUMBERED PARENT



2. NAMING TAIL

Parent: Heptane (7 C), the lower zigzag

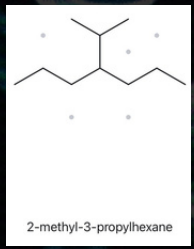
Numbering: Either direction places the branch at C4

Substituent: Propan-2-yl at C4

3. MY NAME

2-(PROPAN-2-YL) HEPTANE

4. THE ALKANE CHECK



Returned name: 2-methyl-3-propylhexane

⊗ Not the same as the given name

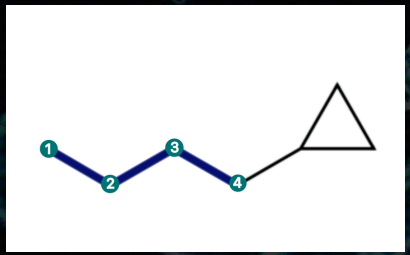
The site selected 6 carbon parent chain, but IUPAC requires choosing thelongest continuous chain, which has 7 carbons.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A10 • 1-CYCLOPROPLYBUTANE

LEVEL: CHALLENGE **MOLECULAR FORMULA: C_7H_{14}**

1. NUMBERED PARENT



2. NAMING TAIL

Parent: Butane (4 C); the attached ring has 3 C, so the chain is parent

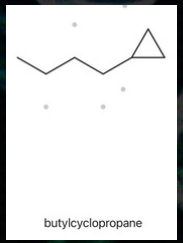
Numbering: Start at the chain end attached to cyclopropyl; locant 1

Substituent: Cyclopropyl at C1

3. MY NAME

1-CYCLOPROPLY BUTANE

4. THE ALKANE CHECK



Returned name: butylcyclopropane

⊗ Not the same as the given name

The site selected 3-carbon ring as the parent, but the attached open chain has 4 carbons. The longer chain takes priority

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD B01 • 2-METHYLPROPANE

LEVEL: FOUNDATION

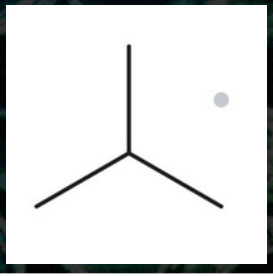
1. NAME DECODED

Parent: Propane; a 3-carbon chain

Substituents: Methyl on C2

Expected formula: 3 + 1 = 4 C; open chain to C_4H_{10}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_4H_{10}

Open chain: $C_nH_{2n+2} = C_4H_{10}$

4. BACK-CHECK

Returned by The Alkane: 2-methylpropane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B02 • 3-ETHYLPENTANE

LEVEL: FOUNDATION

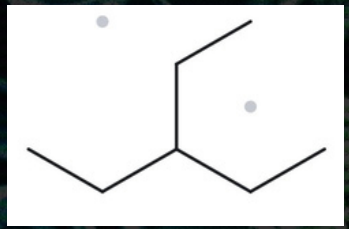
1. NAME DECODED

Parent: Pentane; 5-carbon chain

Substituents: Ethyl on C3

Expected formula: 5+2=7 C; open chain to C_7H_{16}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_7H_{16}

Open chain: CnH_{2n+2} = C_7H_{16}

4. BACK-CHECK

Returned by The Alkane: 3-ethylpentane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B03 • 2,2-DIMETHYLBUTANE

LEVEL: FOUNDATION

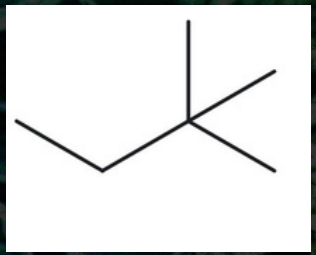
1. NAME DECODED

Parent chain: Butane; 4-Carbon chain

Substituent: Two methyl, both on C2

Expected formula: 4+2=6 C; open chain to C_6H_{14}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_6H_{14}

Open chain: $C_nH_{2n+2} = C_6H_{14}$

4. BACK-CHECK

Returned by The Alkane:
2,2-dimethylbutane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B04 • 3-ETHYL-2-METHYLPENTANE

LEVEL: DEVELOPING

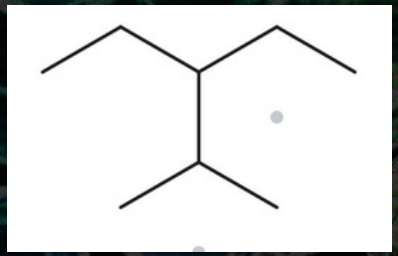
1. NAME DECODED

Parent chain: Pentane; 5-Carbon chain

Substituents: Methyl on C2; ethyl on C3

Expected formula: 5+1+2=8 C; open chain to C_8H_{18}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_8H_{18}

Open chain: $C_nH_{2n+2} = C_8H_{18}$

4. BACK-CHECK

Returned by The Alkane:
3-ethyl-2-methylpentane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B05 • 1-CHLORO-2-METHYL. PROPANE

LEVEL: DEVELOPING

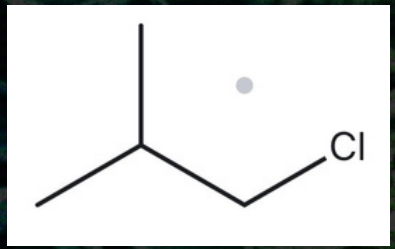
1. NAME DECODED

Parent: propane → a 3-carbon chain.

Substituents: chloro on C1; methyl on C3

Expected formula: 3 + 1 = 4 C; open chain → C_4H_{10}; Cl replaces one H → C_4H_9Cl

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_4H_9Cl

Open chain: C_4H_{10}; minus one H for Cl = C_4H_9Cl

4. BACK-CHECK

Returned by The Alkane:
1-chloro-2-methylpropane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B06 • 1,2-DIMETHYLCYCLOPENTANE

LEVEL: DEVELOPING

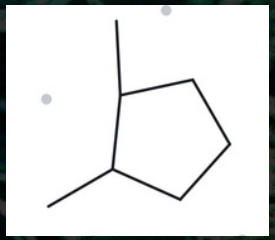
1. NAME DECODED

Parent: cyclopentane → a 5-carbon ring.

Substituents: methyl on C1; methyl on C2.

Expected formula: 5 + 2 = 7 C; one ring → C_7H_{14}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C7H14

One ring: C_nH_{2n} = C_7H_{14}

4. BACK-CHECK

Returned by The Alkane:
1,2-dimethylcyclopentane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B07 • 4-ETHYL-3,3-DIMETYLHEPTANE

LEVEL: DEVELOPING

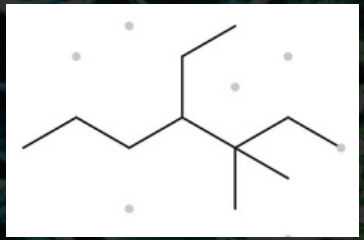
1. NAME DECODED

Parent: heptane → a 7-carbon chain.

Substituents: two methyls on C3; ethyl on C4.

Expected formula: 7 + 2 + 2 = 11 C; open chain → C11H24

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: $C_{11}H_{24}$

Open chain: $C_nH_{2n+2} = C_{11}H_{24}$

4. BACK-CHECK

Returned by The Alkane:
4-ethyl-3,3-dimethylheptane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B08 • TERT-BUTYLCYCLEHEXANE

LEVEL: CHALLENGE

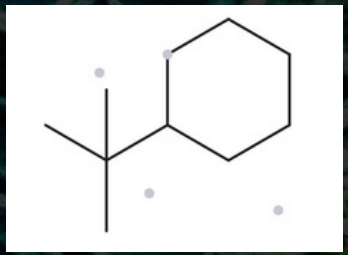
1. NAME DECODED

Parent: cyclohexane → a 6-carbon ring.

Substituents: tert-butyl on C1 (a carbon bearing three methyls, 4 C in total).

Expected formula: 6 + 4 = 10 C; one ring → $C_{10}H_{20}$

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$

One ring: C_nH_{2n} = $C_{10}H_{20}$

4. BACK-CHECK

Returned by The Alkane: (1,1-dimethylethyl)cyclohexane

✓ Same compound, different accepted substituent name. “tert-butyl” is the common name; “1,1-dimethylethyl” is its systematic name (an ethyl with two methyls on its C1).

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B09 • 3-ETHYL-1,1-DIMETHYLCYCLOHEXANE

LEVEL: CHALLENGE

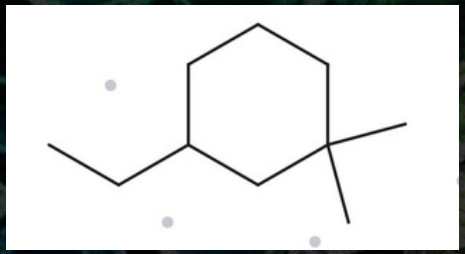
1. NAME DECODED

Parent: cyclohexane → a 6-carbon ring.

Substituents: two methyls on C1; ethyl on C3.

Expected formula: 6 + 2 + 2 = 10 C; one ring → $C_{10}H_{20}$

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$

One ring: C_nH_{2n} = $C_{10}H_{20}$

4. BACK-CHECK

Returned by The Alkane: 3-ethyl-1,1-dimethylcyclohexane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B10 • 2-ETHYLBUTANE

LEVEL: CHALLENGE

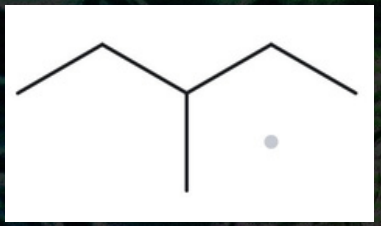
1. NAME DECODED

Parent: Pentane chain → a 5-carbon chain (as given).

Substituents: methyl on C3.

Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}

2. BOND-LINE STRUCTURE (THE ALKANE)



3. FORMULA CHECK

Counted from the drawing: C_6H_{14}

Open chain: $C_nH_{2n+2} = C_6H_{14}$

4. BACK-CHECK

Returned by The Alkane:
3-methylpentane

⊗ IUPAC nomenclature requires selecting the longest continuous carbon chain as the parent. The structure has a five-carbon chain, not four, leaving a methyl substituent at C3. Therefore, the correct name is 3-methylpentane.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded